SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 10: A* Search algorithm

CODE:

```
import heapq

graph = {
    "A": [("B", 1), ("C", 4)],
    "B": [("D", 2), ("E", 5)],
    "C": [("E", 1)],
    "D": [("G", 3)],
    "E": [("G", 2)],
    "G": []
}

h = {"A": 5, "B": 4, "C": 3, "D": 3, "E": 1, "G": 0}

def astar(start, goal):
    pq = [(h[start], 0, start, [start])]
    visited = set()

    while pq:
        f, cost, node, path = heapq.heappop(pq)

        if node == goal:
```

```
            print("Path:", path)
            print("Cost:", cost)
            return

        if node in visited:
            continue
        visited.add(node)

        for neighbor, weight in graph[node]:
            if neighbor not in visited:
                new_cost = cost + weight
                heapq.heappush(
                    pq,
                    (new_cost + h[neighbor],
                     new_cost, neighbor, path + [neighbor])
                )

astar("A", "G")
```

OUTPUT:

